

Self-service checkout potential assessment

A transaction-based decision framework for identifying stores worth SCO consideration

Tina Martinić*

**Human-led. Silicon-backed.*

1. Objective

Identifying stores where SCO is worth further evaluation

The goal is to build a reusable framework for evaluating store-level SCO potential, then apply and validate it using the supplied transaction data. The framework needs to define what makes a good SCO candidate, translate criteria into measurable indicators and explain rules, assumptions, limitations and recommendations.

The objective is not to rank stores by traffic volume alone. A good SCO candidate should combine five dimensions:

Dimension	What it means
Transaction volume	Recurring half-hour periods with enough checkout volume to create queue risk
Basket size	Baskets small enough for SCO to handle efficiently
Time pattern	Pressure that repeats across days and months, not a one-off spike
Staffing implication	SCO should add capacity or replace redundant staffed POS, not remove needed staffed capacity
Store format	Final deployment should fit layout, location, customer mission and operational goals

Core question

Which stores show measurable SCO-suitable pressure?

2. Framework

A gated decision framework for SCO opportunity screening

The framework is designed to identify stores where SCO is worth further evaluation based on measurable transaction evidence.

It uses gates because SCO criteria are not fully compensating. High traffic should not compensate for large baskets, poor data quality, weak recurrence or a staffed POS that is structurally needed.

A good SCO candidate is not the busiest store. It is a store where recurring checkout pressure is small-basket oriented, consistently present and operationally addressable.

Decision flow

Flow	Step	Decision logic
Gate 1	Data quality	remove / flag data issues that could create false pressure or false basket fit
Gate 2	Checkout pressure	identify busy half-hour POS blocks

Gate 3	Small-basket suitability	test whether pressure comes from SCO-suitable baskets
Gate 4	Recurrence and time-pattern	test whether the signal repeats across days / months
Gate 5	Staffed POS logic	decide whether SCO is add-on, replacement, optimization, or not the right intervention
Gate 6	Existing SCO benchmark	use current SCO stores as adoption benchmarks, not as labels
Output	Store classification	2 → strong signal 1 → validate / pilot 0 → defer / diagnose

Why gates instead of weighted scoring?

Weighted scores can rank stores, but they can also hide critical failures. A store with very high traffic can still be a poor SCO candidate if the baskets are large, the data is contaminated by returns or netting, or staffed POS capacity is genuinely needed.

Weighted scores rank opportunities. **Gates prevent bad rollout decisions.**

3. Methodology

From half-hour transaction data to SCO opportunity screening

The methodology applies the framework to the supplied transaction dataset in a structured sequence. The goal is not to infer final deployment readiness, but to determine whether each store shows measurable, recurring and reliable SCO-suitable pressure.

Step	What is calculated	Why it matters
1	Validate and clean input data	Prevent false pressure or false basket fit caused by noisy records
2	Build the operating baseline	Focus the score on regular Mon–Fri operations, excluding Sundays and holidays
3	Aggregate checkout activity	Convert terminal-level rows into store-level half-hour checkout signals
4	Estimate clean basket size	Measure basket suitability without returns, netting or item outliers distorting the result
5	Identify small-basket peak intervals	Find busy half-hour blocks that SCO can realistically absorb
6	Measure recurrence	Check whether the signal repeats across days, months

		and time-of-day patterns
7	Apply staffed POS logic	Decide whether SCO is add-on, replacement, optimization or not the right intervention
8	Classify stores	Assign each store to strong signal, validate / pilot, or defer / diagnose

Key calculation logic

Input data is aggregated at checkout-terminal level by half-hour. Each record contains ticket and item counts for one POS/SCO terminal in one half-hour block, but it does not include individual receipts, product categories, store format, layout, labor schedules or financial inputs. The dataset covers 36 anonymized store IDs, which are treated as store-level analytical units in this analysis.

Each row is identified by:

$$STORE_ID + POS + IS_SELF_CHECKOUT + TIME_BLOCK$$

Rows are first validated, cleaned and enriched with calendar attributes. The core score is then calculated only on regular weekday operations:

- **Monday – Friday**
- **excluding Croatian public holidays and Sundays**

Saturdays are analyzed separately as an interpretation signal, but they do not affect the final score. They can reflect a different shopping context than regular weekdays, so they are useful for store understanding but not for the core baseline.

Sundays and public holidays are excluded because operating patterns are affected by changing regulation and are not a stable basis for SCO screening.

The analysis then converts terminal-level rows into store-level half-hour signals: traffic pressure, clean basket size, clean basket coverage and small-basket peak status.

Core signals	Meaning
POS tickets	Checkout volume in a store-level half-hour block
clean basket tickets	Tickets that can be safely used for basket-size scoring
clean basket items	Items from rows that are not affected by returns, netting or basket outliers
clean items per ticket	Basket size after excluding noisy basket rows
clean basket coverage	Share of tickets in the block that are reliable enough for basket scoring
early checkout pressure	Whether the block has enough traffic to matter operationally

small-basket peak status	Whether the block is both busy and SCO-suitable
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A **half-hour block** is counted as a **strong small-basket pressure interval** only if it passes all three conditions:

- i) *POS tickets* ≥ 30
- ii) *AND clean basket coverage* $\geq 70\%$
- iii) *AND clean items per ticket* ≤ 4

A broader pilot signal uses the same traffic and coverage requirements, but allows clean items per ticket up to 5.

The block-level signals are then aggregated into store-level indicators that measure both signal size and recurrence:

Store-level indicator	Interpretation
Number of small-basket peak intervals	Absolute volume of SCO-suitable pressure
Small-basket peaks per 100 observed open half-hours	Normalized pressure intensity
Share of baseline days with at least one small-basket peak	How often the pressure appears across operating days
Number of months with recurring pressure	Whether the signal is stable over time

This prevents the recommendation from being driven by a single strong day, an isolated spike or a short-lived anomaly.

Final screening output

The final score is assigned by evaluating the store-level indicators against the framework gates and thresholds:

Score	Interpretation
2	Strong transaction-based signal; store is worth serious SCO evaluation
1	Moderate SCO-suitable signal; pilot / field validation is needed
0	Defer / diagnose; not enough clean SCO-suitable evidence

This is a **screening output**, not final deployment approval. It identifies where SCO is worth further evaluation based on transaction evidence. Final deployment still requires validation of store format, layout, labor implications, CAPEX/OPEX, margin, payback and potential retail-media upside.

4. Data-quality safeguards

The framework is strict because noisy data can create false SCO candidates

Data quality is not a side check. It is part of the decision logic. Noisy records can distort either checkout pressure or basket suitability, which could create false SCO candidates.

Issue / treatment area	Treatment
Exact duplicate rows	Removed before scoring
Traffic outlier >70 tickets / terminal / 30 min	Removed before scoring as physically implausible traffic
Positive items with zero tickets	Removed before scoring
NUMBER_OF_ITEMS < NUMBER_OF_TICKETS	Kept as traffic, excluded from basket scoring
Negative items	Treated as return/correction workload, not as SCO basket demand
Basket-size outliers	Tickets kept as workload signal; items excluded from basket scoring
Duplicate-key rows	Measured by ticket share; material cases reduce data-quality confidence
Croatian public holidays	Excluded from core baseline
Saturdays	Excluded from core baseline as weekend shopping patterns can differ from weekday operations
Sundays	Excluded because operating patterns are affected by regulation and are not a stable SCO-screening baseline

Core principle:

Bad ticket data can create false pressure.

Bad item data can create false basket fit.

The framework protects against both.

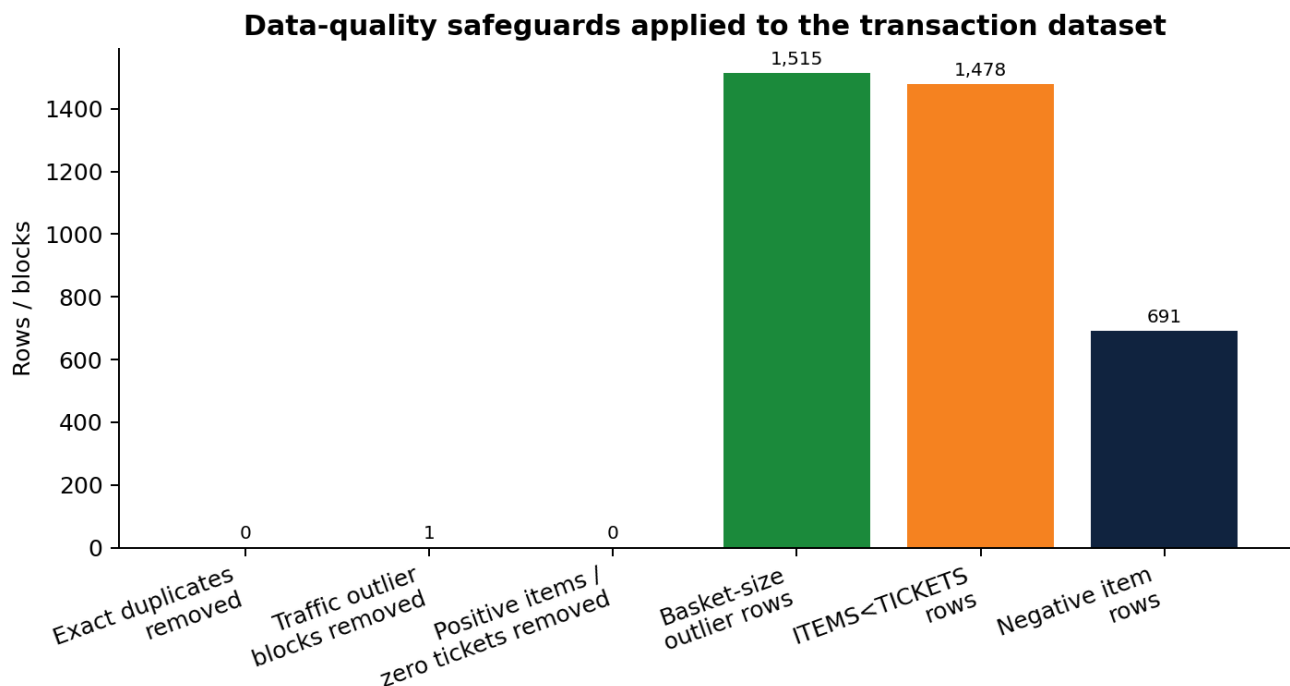


Figure 1. Data-quality safeguards applied before scoring.

The chart shows how many rows or terminal half-hour blocks were removed, flagged or excluded from basket-size scoring to prevent false checkout pressure or false small-basket fit. Traffic outliers and invalid zero-ticket item rows are removed before scoring, while basket-size outliers, ITEMS < TICKETS rows and negative-item rows are retained as workload signals but excluded from clean basket scoring where relevant.

5. Scoring rules and thresholds

The framework uses explicit, adjustable decision rules

The score is not calculated as a weighted average. Each store is evaluated through a sequence of rules that test whether checkout pressure is strong, small-basket oriented, recurring and reliable enough to support SCO opportunity screening.

The thresholds below define the default screening setup used in the analysis.

Rule	Default threshold	Purpose
Early checkout pressure	≥ 30 POS tickets / 30 min	Identifies half-hour blocks with enough checkout volume to matter operationally
Capacity breach reference	$80\% \times 1800 \text{ sec} / (23 + 3 \times \text{estimated basket size})$	Store-specific supporting metric for staffed-POS pressure interpretation
Strong basket fit	≤ 4 clean items / ticket	Conservative upper bound for strong small-basket screening signal
Pilot basket fit	≤ 5 clean items / ticket	Broader basket threshold for validation cases
Clean basket coverage	$\geq 70\%$	Ensures the basket signal is reliable enough to use

Strong small-basket pressure	≥100 intervals, ≥3 per 100 observed open HH, ≥45% baseline days	Identifies recurring and material SCO-suitable pressure
Pilot / validation pressure	≥50 intervals, ≥1.5 per 100 observed open HH, ≥20% baseline days	Identifies moderate SCO-suitable signal requiring validation
Monthly consistency	≥4 months passing monthly pressure thresholds	Reduces the risk of decisions driven by one-off spikes or short-lived anomalies
High netting risk	≥10% tickets in ITEMS < TICKETS rows	Triggers low data-quality confidence
High return/correction share	≥2% tickets in negative-item rows	Triggers low data-quality confidence
High duplicate-key risk	≥10% tickets in duplicate-key rows	Triggers low data-quality confidence
Traffic outlier	>70 tickets / terminal / 30 min	Removed from scoring as physically implausible traffic
True multi-POS usage	second POS ≥5 tickets, ≥30% share, total POS tickets ≥20	Tests whether the additional staffed POS is materially used

The capacity-breach reference is a secondary operational signal. It estimates when staffed-POS workload may approach practical capacity using a service-time assumption of 23 seconds per ticket + 3 seconds per item, applied at 80% utilization over a 30-minute block. The 80% assumption keeps the reference below theoretical maximum capacity, allowing for checkout variability, staff interruptions and non-scanning tasks. This metric supports interpretation; the final score is still driven by recurring small-basket pressure, data quality and POS setup.

The ≤4 clean items per ticket threshold is used as a conservative upper bound for strong small-basket pressure. It is aligned with the provided SCO context, where existing SCO usage is presented around four items per basket, and validated directionally against the analyzed data, where existing SCO baskets are even smaller than staffed POS baskets.

The framework treats **thresholds as configurable parameters**, allowing sensitivity testing without changing the underlying decision logic.

6. Staffing and POS intervention logic

The framework never removes the last staffed POS

Staffing is a key part of the screening logic. SCO should not remove staffed capacity that is structurally needed, especially where the store has only one staffed POS available for cash, returns, assistance, larger baskets and fallback situations.

POS setup rules:

Store setup	Framework logic	Business interpretation
One staffed POS, no SCO	Keep staffed POS; SCO can only be added	The last staffed POS is preserved
One staffed POS + existing SCO	Treat as optimization / adoption case	Existing SCO usage should be benchmarked, not treated as replacement
Two or more staffed POS terminals	Test whether the additional POS is materially used in high-pressure intervals	Replacement is only considered if the additional staffed POS appears redundant

True multi-POS usage is identified when all above conditions are met:

- i) *active staffed POS terminals* ≥ 2
- ii) *AND second – strongest POS tickets* ≥ 5
- iii) *AND second – strongest POS share* $\geq 30\%$
- iv) *AND total POS tickets* ≥ 20

Replacement interpretation

Additional staffed POS assessment	SCO-suitable pressure	Framework interpretation
Structurally needed	Any	Keep staffed POS; SCO is not a replacement case
Redundant	Present and recurring	SCO / hybrid replacement can be considered
Redundant	Weak or absent	Repurpose may be considered, but not as an SCO case
Unclear	Unclear or data-limited	Field validation required

Business interpretation

Redundant POS $\neq$ automatically SCO candidate

SCO replacement case

$$= \text{redundant staffed POS} + \text{measurable SCO} - \text{suitable pressure} + \text{acceptable data quality}$$

This separates two decisions that are often confused: whether additional staffed POS capacity is needed, and whether SCO is the right way to use or replace that capacity.

7. Findings and recommendations

Recommended action depends on pressure type, existing SCO and POS setup

The framework classified the 36 store IDs available in the analyzed transaction dataset into three screening groups:

Classification	Stores	Interpretation
Score 2	6	Strong transaction-based signal; worth serious SCO evaluation
Score 1	1	Moderate SCO-suitable signal; pilot / field validation needed
Score 0	29	Defer; not enough clean SCO-suitable evidence

Main candidates:

Store	Score	Recommendation	Why
18	2	Confirm / optimize existing SCO	strongest signal; existing SCO should be optimized
4	2	Add SCO candidate	strong signal, no SCO, one staffed POS retained
35	2	Confirm / optimize existing SCO	strong signal and existing SCO
14	2	Confirm / optimize existing SCO	strong signal and existing SCO
29	2	Add SCO candidate	strong signal, no SCO, one staffed POS retained
25	2	Replace redundant POS with SCO / hybrid	redundant POS + SCO-suitable pressure
30	1	Pilot / validate POS replacement	moderate SCO-suitable signal, below strong-signal threshold

Threshold robustness

To test whether the Score 2 list is driven by underlying patterns or by where the thresholds are set, the three core strong-signal thresholds were tested under $\pm 20\%$ perturbations applied simultaneously:

Threshold	Default	Looser setup	Stricter setup
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Minimum strong small-basket peak intervals	100	80	120
Minimum peaks per 100 observed open half-hours	3.0	2.4	3.6
Minimum baseline day share	45%	36%	54%

Across all three setups, the Score 2 group remained unchanged: stores 4, 14, 18, 25, 29 and 35. The Score 1 / Score 0 split also remained unchanged at 1 / 29.

This indicates that the strong-signal recommendation is not driven by precise calibration of the final recurrence thresholds. The test does not claim full robustness to all signal-definition parameters. Earlier thresholds, such as early checkout pressure, basket-fit threshold and clean basket coverage, define what the framework counts as SCO-suitable pressure and should remain configurable for context-specific tuning.

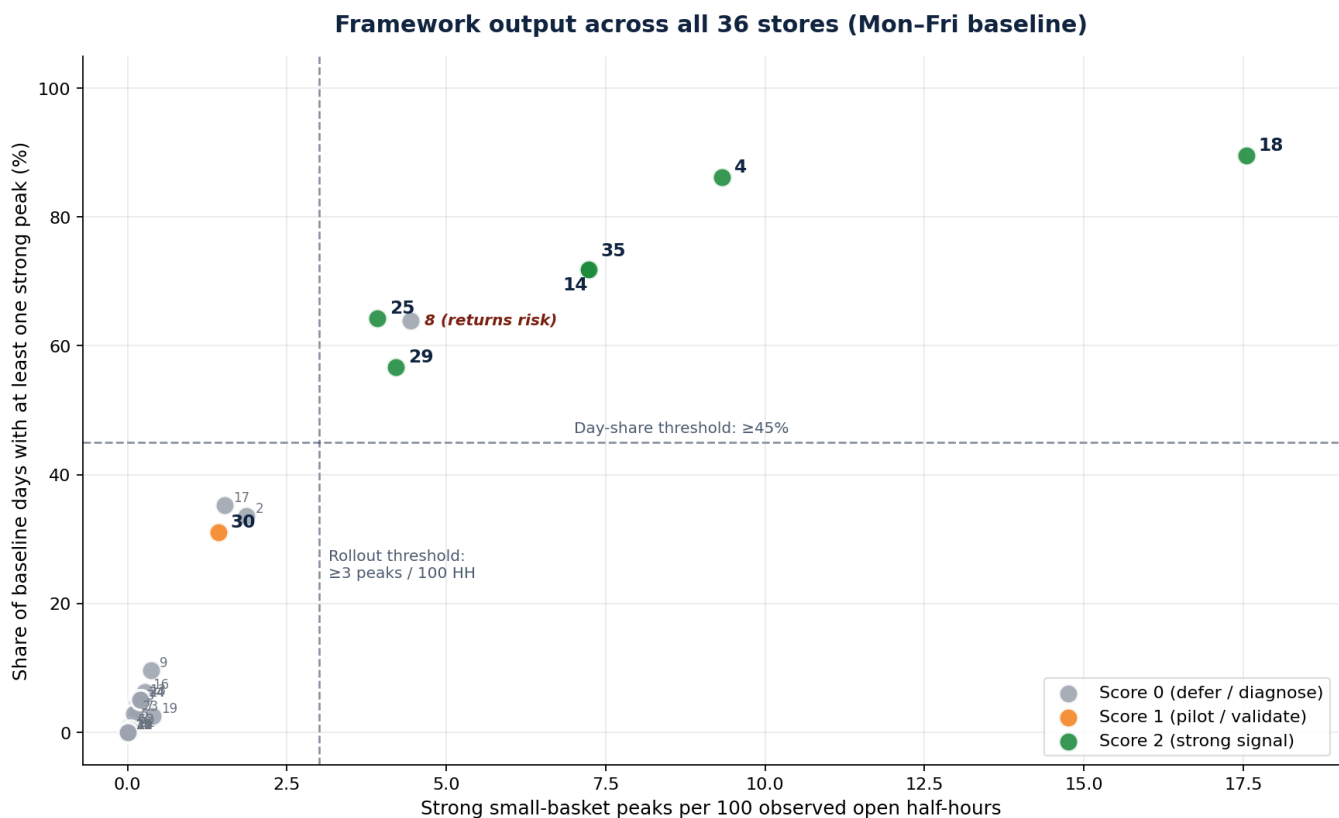


Figure 2. Framework output across all analyzed store IDs.

Stores are plotted by normalized strong small-basket pressure and share of baseline days with at least one strong peak. Green stores meet the strong-signal criteria, orange indicates pilot / validation, and grey stores are deferred. Store 8 illustrates the value of the data-quality gate: it shows strong pressure, but is excluded from recommendation due to returns / netting risk.

Why selected score-0 stores are deferred?

Not all score-0 stores are weak for the same reason. Two diagnostic examples show why the framework does not rely on traffic alone.

Store	What the data shows	Why it is score 0?	Interpretation
8	Strong small-basket pressure and potentially	High netting risk and high return/correction	Looks like an SCO candidate at first, but the signal is not

	redundant staffed POS	share reduce data confidence	reliable enough for recommendation
19	Clean data, but only 13 strong small-basket peak intervals and 0.39 peaks per 100 observed open half-hours	Signal remains weak even after normalizing by observed open half-hours	Defer is not caused by shorter operating coverage; SCO-suitable pressure is simply not recurring enough

Store 8 demonstrates the value of the data-quality gate: apparent pressure should not become an SCO recommendation if the signal may be distorted by returns, corrections or netting.

Store 19 demonstrates the value of normalizing by observed open half-hours: the framework does not penalize potentially shorter operating coverage, but the signal remains too weak even after normalization.

Defer / diagnose group

The remaining 29 stores do not show enough clean, recurring SCO-suitable pressure to justify SCO as the primary intervention based on transaction evidence.

This does not necessarily mean that these stores have no operational issue. Some may still have checkout workload, but the data does not support SCO as the right answer. In those cases, the next step is operational diagnosis rather than capex.

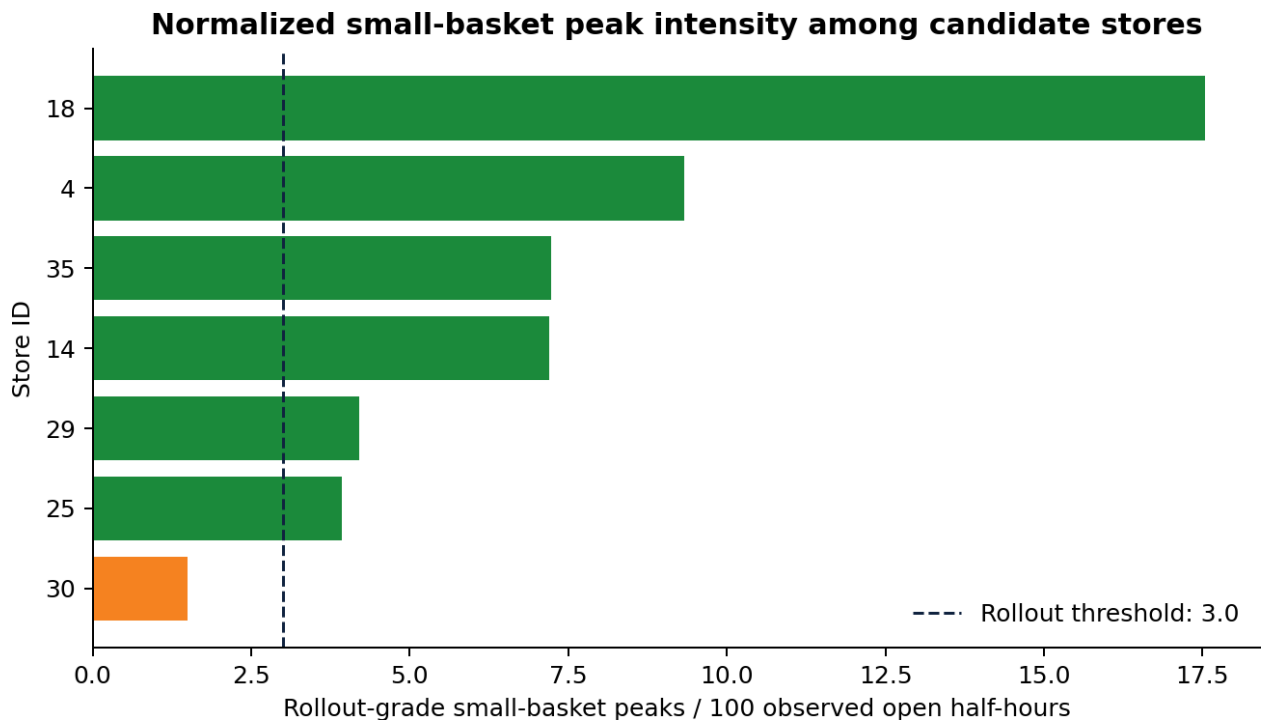


Figure 3. **Candidate stores ranked by normalized small-basket peak intensity.**

Green bars represent strong-signal stores, while the orange bar indicates a pilot / validation case. The dashed line marks the strong-signal threshold of 3 peaks per 100 observed open half-hours.

8. Existing SCO benchmark

Existing SCO is used as a benchmark, not as a label

Existing SCO presence is not used as a target label or proof of optimal deployment. Instead, stores with existing SCO are used to benchmark actual usage behavior: adoption, basket separation and peak-period usage.

Metric	Meaning
sco_ticket_share	Share of checkout tickets processed through SCO
POS items/ticket – SCO items/ticket	Whether SCO is used for smaller baskets than staffed POS terminals
Review flags	Low adoption, weak basket separation or lower SCO usage during peak periods

Good existing SCO pattern

high SCO share + positive basket separation + relevant pressure + no review flags

This answers a different question than rollout screening:

When SCO already exists, is it being used for the intended small-basket mission?

Applied to the analyzed data

In the analyzed transaction dataset, four stores already have SCO terminals: Stores 13, 14, 18 and 35. These stores are benchmarked to understand whether existing SCO usage follows the expected pattern: meaningful adoption, smaller baskets than staffed POS and relevant use during checkout pressure.

Low adoption is flagged when SCO ticket share is below 12% while weak basket separation is flagged when POS-SCO basket gap is below 0.5 items/ticket.

The benchmark shows that Stores 18, 35 and 14 have a clean existing-SCO pattern, with no review flags and clear basket separation versus staffed POS.

Store 13 shows good basket separation, but requires review because SCO adoption is below the low-adoption threshold.

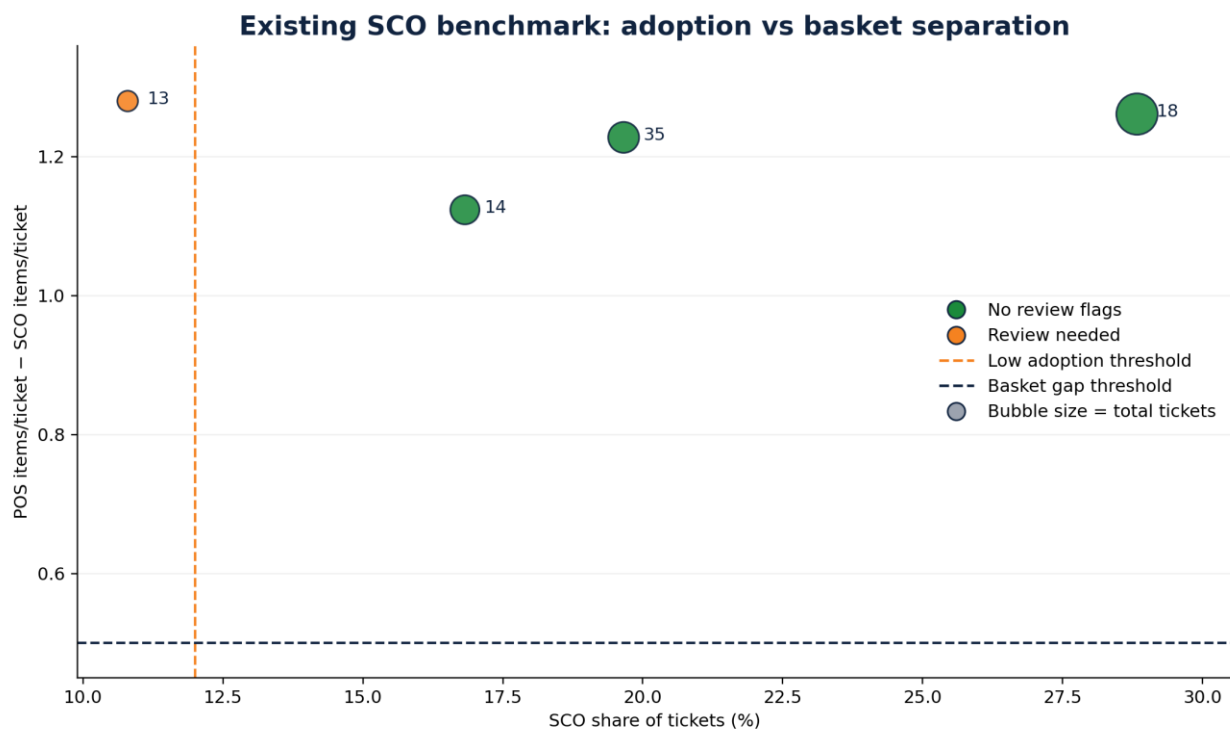


Figure 4. Existing SCO benchmark across stores that already have SCO.

9. Limitations and next data needed

Transaction evidence is not the same as deployment readiness

The framework is intentionally transaction-based. It screens for stores where the data shows recurring, small-basket checkout pressure that could justify further SCO evaluation. It does not approve final deployment.

The transaction dataset used in this analysis is synthetic. Store-level outputs should therefore be read as framework results under the available data, not as literal real-world deployment decisions.

The framework also stays within what the transaction data can support. It does not infer store format, layout feasibility, product mission, labor impact or financial return from fields that are not present in the dataset.

Key limitations

Limitation	Why it matters
Synthetic transaction dataset	Store-level results demonstrate the framework logic, but should not be read as literal real-world deployment decisions
No store-format data	Urban, tourist, residential or transit context can materially affect SCO adoption and mission fit
No layout / selling-area data	A store may show SCO-suitable pressure but still lack feasible checkout-zone space
No product category or mission data	Basket size is used as a proxy, but category and mission data would better confirm impulse, top-up or fresh missions
No cash share / service	Cash usage, parcel services, returns and staff-only tasks can

workload data	affect whether staffed POS capacity must remain
No financial inputs	CAPEX, maintenance, margin, labor cost, average ticket value and lost-sales estimates are needed for payback
No lost-sales / queue-abandonment data	Checkout pressure shows where queue risk may exist, but not how much revenue is actually lost or protected by reducing wait time
No retail-media inputs	Advertising-enabled SCO requires screen visibility, supplier/media demand and commercial-rate assumptions

Missing half-hour coverage

Missing half-hour blocks are treated as non-observed periods, not as open periods with zero tickets. This avoids assuming store opening hours that are not provided.

If blocks are missing because the store was closed, renovated or operating with different hours, this treatment is appropriate. **If blocks are missing due to export error, pressure metrics may be under- or over-estimated.**

10. Next steps

From opportunity screening to deployment decision

Once a store shows a qualifying SCO signal, the next step is to validate whether deployment is physically, operationally and financially justified.

Data layer	Required inputs	How it upgrades the framework
Store format	urban/tourist/residential/transit context, micro-location	Confirms whether the customer mission fits SCO
Layout feasibility	selling area, checkout-zone space, queue flow, power/network readiness	Determines whether SCO can physically fit
Product/category mission	product categories, promo flags, restricted items, fresh/weighed items	Replaces basket-size proxy with stronger mission evidence
Labor and operations	staffing schedules, task allocation, SCO intervention rate, cash/service workload	Tests whether SCO frees capacity or creates extra work
Financial case	CAPEX, OPEX, margin, labor cost, average ticket value, lost-sales / queue-abandonment estimate	Calculates store-level payback and quantifies whether reducing checkout friction creates enough business value

Retail media potential	screen visibility, expected exposure, supplier demand, media rates	Quantifies additional upside from advertising-enabled SCO
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Payback as a separate validation layer

Payback should be calculated with validated internal assumptions, including CAPEX, maintenance, gross margin, labor cost, average ticket value, queue-related lost-sales estimate and potential retail-media revenue.

Retail media upside

Retail media should be treated as an additional business-case layer, not as part of the transaction-based score. Once a store shows qualifying SCO pressure, existing or planned SCO traffic can be translated into potential media inventory.

$$\text{Annual retail media impressions} = \text{SCO transactions} \times \text{eligible screen exposures} \times \text{campaign sell-through}$$

$$\text{Annual retail media revenue} = \text{annual retail media impressions} \times \text{CPM} / 1,000$$

Parameter	Meaning
SCO transactions	Number of checkout transactions processed through SCO in a given period
Eligible screen exposures	Number of ad-viewable screen opportunities per SCO transaction, depending on checkout flow, screen placement and dwell time
Campaign sell-through	Share of available retail-media inventory that is actually sold to campaigns
CPM	Revenue per 1,000 ad impressions
Annual retail media impressions	Estimated yearly volume of monetizable ad impressions on SCO screens
Annual retail media revenue	Potential yearly retail-media revenue generated from those impressions

The transaction data can indicate where SCO traffic exists or could grow, but it cannot estimate retail-media revenue on its own. That requires additional inputs: screen visibility, exposure logic, campaign sell-through, supplier demand and commercial CPM. Stores with strong existing SCO usage, such as Store 18, are natural candidates for this next-step upside test.

Final logic

Transaction data identifies where SCO is worth further evaluation.
Operational data confirms whether deployment is feasible.
Financial data confirms whether deployment is justified.

11. Closing recommendation

Use the framework as an SCO opportunity screening tool

The framework should be used as the first decision layer before store-level deployment planning. It separates stores with a qualifying transaction-based SCO signal from stores that require further diagnosis or different operational interventions.

Recommended use

Step	Purpose
1. Screen stores using transaction evidence	Identify stores with recurring checkout pressure, small clean baskets, reliable data and feasible staffed-POS setup
2. Validate operational deployment	Confirm store format, layout, customer mission, cash/service workload, staff availability and existing SCO adoption if relevant
3. Add the financial case	Add CAPEX/OPEX, margin, labor redeployment, average ticket value, queue-related lost-sales estimate, payback and potential retail-media upside

Final message

SCO should not be evaluated only where traffic is highest. It should be evaluated where recurring small-basket pressure is present, where staffed POS capacity can be preserved or reconfigured safely, and where operational and financial validation can confirm deployment readiness.

Reusable by design

New data can be uploaded, thresholds can be adjusted, and the same logic can evaluate future stores without relying on the current SCO footprint.

Supporting artifact

The [Streamlit app](#) operationalizes the framework, supports threshold sensitivity testing and provides store-level deep dives.
Streamlit app available via QR:

